

Diverse Sources of C. difficile Infection Identified on Whole-Genome Sequencing



BACKGROUND It has been thought that *Clostridium difficile* infection is transmitted predominantly within health care settings. However, endemic spread has hampered identification of precise sources of infection and the assessment of the efficacy of interventions.

METHODS From September 2007 through March 2011, we performed whole-genome sequencing on isolates obtained from all symptomatic patients with *C. difficile* infection identified in health care settings or in the community in Oxfordshire, United Kingdom. We compared single-nucleotide variants (SNVs) between the isolates, using *C. difficile* evolution rates estimated on the basis of the first and last samples obtained from each of 145 patients, with 0 to 2 SNVs expected between transmitted isolates obtained less than 124 days apart, on the basis of a 95% prediction interval. We then identified plausible epidemiologic links among genetically related cases from data on hospital admissions and community location.

RESULTS Of 1250 *C. difficile* cases that were evaluated, 1223 (98%) were successfully sequenced. In a comparison of 957 samples obtained from April 2008 through March 2011 with those obtained from September 2007 onward, a total of 333 isolates (35%) had no more than 2 SNVs from at least 1 earlier case, and 428 isolates (45%) had more than 10 SNVs from all previous cases. Reductions in incidence over time were similar in the two groups, a finding that suggests an effect of interventions targeting the transition from exposure to disease. Of the 333 patients with no more than 2 SNVs (consistent with transmission), 126 patients (38%) had close hospital contact with another patient, and 120 patients (36%) had no hospital or community contact with another patient. Distinct subtypes of infection continued to be identified throughout the study, which suggests a considerable reservoir of *C. difficile*.

CONCLUSIONS Over a 3-year period, 45% of *C. difficile* cases in Oxfordshire were genetically distinct from all previous cases. Genetically diverse sources, in addition to symptomatic patients, play a major part in *C. difficile* transmission. (Funded by the U.K. Clinical Research Collaboration Translational Infection Research Initiative and others.)